Phase Plane Analysis and Nonlinear Dynamics

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from classifying a linear equilibrium by trace and determinant, through building the Jacobian of a nonlinear system, finding fixed points by nullclines, and linearizing the pendulum, up to a full Lotka–Volterra predator–prey analysis with the Jacobian evaluated and the eigenvalues classified at every equilibrium. Solutions follow in Part III.

1. **(Classify a linear system)** For $A = \begin{pmatrix} -1 & 2 \\ 0 & -4 \end{pmatrix}$, compute τ and Δ and classify the origin of $\mathbf{x}' = A\mathbf{x}$.
2. **(Spiral or node?)** For $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, find the eigenvalues and classify the origin.
3. **(Eigendirections)** For $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$, find the eigenvalues and eigenvectors and state the equilibrium type.
4. **(Build the Jacobian)** For the system $x' = x - xy,\ y' = -y + xy$, write the Jacobian matrix $J(x, y)$ symbolically.
5. **(Fixed points by nullclines)** Find all fixed points of $x' = x(3 - x - 2y),\ y' = y(2 - x - y)$.
6. **(Linearize the pendulum)** For $x' = y,\ y' = -\sin x$, compute $J(x, y)$ and evaluate it at the fixed points $(0, 0)$ and $(\pi, 0)$.
7. **(Classify the pendulum equilibria)** Using the matrices from Problem 6, find the eigenvalues at $(0, 0)$ and $(\pi, 0)$ and classify each.
8. **(Predator–prey fixed points)** For $x' = x(2 - y),\ y' = y(-3 + x)$, find both equilibria.
9. **(Predator–prey Jacobian)** For the system in Problem 8, evaluate the Jacobian at the coexistence equilibrium and classify it.
10. **(Nonlinear capstone)** For $x' = y,\ y' = -x + x^3$, find all fixed points, evaluate the Jacobian at each, classify them, and describe the global phase portrait.

Part III — Complete Solutions

Solution 1. Trace and determinant give the type directly:

$$\tau = \operatorname{tr} A = -5, \quad \Delta = \det A = (-1)(-4) - 0 = 4 > 0, \quad \tau^2 - 4\Delta = 25 - 16 = 9 > 0.$$

With $\Delta > 0$, discriminant > 0, and $\tau < 0$, the origin is a **stable node** (eigenvalues $-1, -4$).

Solution 2. Here $\tau = 0$ and $\Delta = 0 \cdot 0 - (-2)(2) = 4 > 0$:

$$\det(A - \lambda I) = \lambda^2 + 4 = 0 \;\Rightarrow\; \lambda = \pm 2i.$$

Pure imaginary eigenvalues with $\tau = 0$ make the origin a **center** — closed orbits around the origin.

Solution 3. Expand the characteristic polynomial and solve:

$$\det(A - \lambda I) = (1-\lambda)^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) \Rightarrow \lambda_1 = 3,\ \lambda_2 = -1.$$

$$\lambda_1 = 3 :\ \mathbf{v}_1 = (1, 2); \qquad \lambda_2 = -1 :\ \mathbf{v}_2 = (1, -2).$$

Opposite signs $\Rightarrow$ a **saddle**: out along $(1, 2)$, in along $(1, -2)$.

Solution 4. Here $f = x - xy$ and $g = -y + xy$, so the four partials are

$$J(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} = \begin{pmatrix} 1 - y & -x \\ y & -1 + x \end{pmatrix}.$$

Solution 5. Factor each velocity. The x-nullclines are $x = 0$ or $3 - x - 2y = 0$; the y-nullclines are $y = 0$ or $2 - x - y = 0$. Pairing branches:

$$(0,0), \quad (0,2), \quad (3,0), \quad \text{and } 3 - x - 2y = 0,\ 2 - x - y = 0 \Rightarrow (1,1).$$

Four fixed points: $(0,0), (0,2), (3,0), (1,1)$.

Solution 6. With $f = y,\ g = -\sin x$:

$$J(x,y) = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}; \quad J(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J(\pi, 0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(Using $\cos 0 = 1$ and $\cos \pi = -1$.)

Solution 7. Take the characteristic polynomial of each:

$$(0,0) :\ \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i \text{ (center)}; \qquad (\pi, 0) :\ \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \text{ (saddle)}.$$

The hanging-down equilibrium oscillates (center); the inverted equilibrium is an unstable saddle.

Solution 8. Factor: $x(2 - y) = 0 \Rightarrow x = 0$ or $y = 2$; $y(-3 + x) = 0 \Rightarrow y = 0$ or $x = 3$. Pairing the branches,

$$(0,0) \text{ (extinction)} \qquad \text{and} \qquad (3,2) \text{ (coexistence)}.$$

Solution 9. Here $f = 2x - xy,\ g = -3y + xy$, so $J = \left(\begin{smallmatrix} 2-y & -x \\ y & -3+x \end{smallmatrix}\right)$. At the coexistence point $(3, 2)$,

$$J(3,2) = \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix}, \quad \det(J - \lambda I) = \lambda^2 + 6 = 0 \Rightarrow \lambda = \pm i\sqrt{6}.$$

Pure imaginary eigenvalues $\Rightarrow$ a **center**: predator and prey populations cycle around $(3, 2)$.

Solution 10. Fixed points: $y = 0$ and $-x + x^3 = x(x^2 - 1) = 0 \Rightarrow (0, 0), (\pm 1, 0)$. The Jacobian is $J = \left(\begin{smallmatrix} 0 & 1 \\ -1+3x^2 & 0 \end{smallmatrix}\right)$:

$$J(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \Rightarrow \lambda = \pm i \text{ (center)}; \quad J(\pm 1, 0) = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \Rightarrow \lambda = \pm\sqrt{2} \text{ (saddles)}.$$

Globally: closed orbits around the origin, hemmed in by the separatrices joining the two saddles at $(\pm 1, 0)$ — a conservative double-well picture.

Unit Key Takeaway

One geometric idea runs through the unit: classify the flow one equilibrium at a time. For a linear system the trace and determinant of A name the origin outright. For a nonlinear system, locate the fixed points as nullcline intersections, then linearize with the Jacobian $J = \left(\begin{smallmatrix} f_x & f_y \\ g_x & g_y \end{smallmatrix}\right)$ — the best linear approximation — and classify each fixed point by its eigenvalues. Stitching those local portraits together along the flow arrows reconstructs the global dynamics, from the swinging pendulum to the cycling predator–prey populations.